AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT132
Linear Algebra

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 20/Nov./2025

Number of Exam Pages: 2 pages (including this cover sheet)	**Time Allowed:** 90 Minutes

Instructions:

- Total Marks: **30**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Q–1: [4×2 marks] Choose the correct answer:

1.1: If A is an invertible matrix, the solution to the linear system $AX = B$ is

a) $X = B - A$
b) $X = BA^{-1}$
c) $X = A^{-1}B$
d) $X = B^{-1}A$
e) None of the above

1.2: Which of the following matrices does not have an inverse?

a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
b) $\begin{bmatrix} 0 & 5 \\ 1 & 0 \end{bmatrix}$
c) $\begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$
d) $\begin{bmatrix} 2 & -2 \\ 3 & -3 \end{bmatrix}$
e) None of the above

1.3: The determinant of the identity matrix of the size n is

a) n
b) 0
c) 1
d) -1
e) None of the above

1.4: Matrix A is invertible and $\det(A) = 2$. What is $|-3A|$?

a) 216
b) -216
c) 54
d) -54
e) None of the above

Q–2: [8 marks] Answer the following questions as True or False (1 mark), Justifying your answer (3 marks):

2.1: The sum of two symmetric matrices is always symmetric matrix.

2.2: If $\begin{vmatrix} a & 1 \\ 0 & 3 \end{vmatrix} = \begin{vmatrix} 1 & a \\ 2 & 0 \end{vmatrix} + \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix}$, then $a = 0$.

Each question is worth 7 marks. Answer the following questions:

Q–3 [5+3 marks]: Consider the linear system: $\begin{cases} x + 2y + z = 0 \\ x + y + z = 2 \\ 2x + y + z = 1 \end{cases}$

a) Find the coefficient matrix of the linear system, A, and find its inverse.
b) Solve the linear system.
c) Is the linear system inconsistent? Justify.

Q–4 [4+3 marks]: Let $A = \begin{bmatrix} -1 & 0 & 0 & 2 \\ 0 & -2 & 2 & 0 \\ 3 & 0 & 0 & 6 \\ 0 & 1 & 5 & 5 \end{bmatrix}$

a) Find $|A|$
b) Find $\left|3A^{-1}A^2\right|$.
c) Is the matrix A invertible? Justify.

End of questions

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